

Local Android Device Environment (100% Offline — No android.permission.INTERNET)

PRESENTATION LAYER (Jetpack Compose)

12-Hour Grid Time Picker

- 56x56dp Touch Buttons
- AM/PM + Minute Chips

High-Contrast UI Themes

- WCAG AAA (7.8:1 Ratio)
- Scalable Text up to 200%

Full-Screen Lock Overlay

- FLAG_SHOW_WHEN_LOCKED
- One-Tap Take / Snooze

DOMAIN LAYER (Business Logic & Scheduling Engine)

Adherence State Machine

- TAKEN, SCHEDULED
- LATE & MISSED Windows
- PDC Calculation Engine

Exact Alarm Subsystem

- AlarmManager (RTC_WAKEUP)
- SCHEDULE_EXACT_ALARM
- Boot Auto-Reschedule

Offline ISO Emergency Resolver

- SIM / Network ISO Code
- Local Country Number Map
- Intent.ACTION_DIAL Dispatch

DATA & STORAGE LAYER (Local-Only Persistence)

Room SQLite Database

- User Entity (Settings)
- Medicine Entity (Schedules)
- ScheduleEntry Entity (Logs)

Jetpack DataStore

- Preferences DataStore
- Theme & Font Settings
- Sound & Vibration Profiles

Native PDF Generator

- Client-Side PdfDocument
- Standard 595x842 pt A4
- Local Storage / Share Sheet

ANDROID OS KERNEL & SECURITY BOUNDARY

No android.permission.INTERNET ==> Linux Kernel Blocks All AF_INET / AF_INET6 Sockets
All Patient Medication Records Remain Exclusively on Local Hardware Storage